

Experiment: Extraction and Characterization of Gelatin from Delimed Tannery Trimmings

Acid-assisted thermal extraction using acetic acid — a group-based experiment

1. Introduction & Background Concepts

The leather industry turns hide into leather but discards large amounts of collagen-rich solid waste — trimmings, splits, fleshings and shavings. Non-chrome trimmings from the limed/delimed (beamhouse) stage are almost pure collagen and are therefore a clean, valuable raw material. Converting them into gelatin is a classic example of waste valorisation and circular-economy practice: it removes a disposal problem and creates a high-value product. This experiment extracts gelatin from such trimmings and measures its quality.

What is collagen?

Collagen is the main structural protein of skin and hide (about a third of all animal protein). Three polypeptide chains wind around one another into a rope-like triple helix (tropocollagen), held together by hydrogen bonds and covalent cross-links. These triple helices pack into fibrils and fibres that give hide its strength. Collagen is unusually rich in the amino acids glycine, proline and hydroxyproline.

What is gelatin, and how is it formed?

Gelatin is the water-soluble protein obtained when collagen is partly broken down. When the cross-links and hydrogen bonds are loosened (by acid or alkali) and the material is heated, the ordered triple helix unwinds into separate, soluble random-coil chains — this is gelatin. On cooling, these chains partly re-form helices and trap water, giving the familiar gel. The change collagen → gelatin is mainly a denaturation (unfolding), with some hydrolysis (breaking of peptide bonds); too much hydrolysis lowers the molecular weight and weakens the gel.

Why acetic acid?

Acetic acid penetrates and swells the collagen fibres by disrupting the ionic (charge–charge) interactions between chains, without dissolving the protein or hydrolysing it heavily. This swelling opens the structure so that, on heating, the triple helix converts smoothly to soluble gelatin. Acid-processed collagen yields “Type A”-character gelatin (as opposed to “Type B” from alkaline/lime processing).

2. Objective

To extract gelatin from delimed (non-chrome) tannery trimmings by acid-assisted thermal hydrolysis using acetic acid, and to determine the yield and the physicochemical properties of the recovered gelatin. As a class, to find how the extraction temperature affects yield and gel quality.

3. Principle

Leather trimmings from limed hides consist mainly of collagen. During acid pre-treatment, acetic acid penetrates and swells the collagen fibres by disrupting intermolecular ionic interactions, without extensive hydrolysis. Subsequent heating converts the triple-helical collagen into soluble gelatin through partial denaturation. Insoluble residues are removed by filtration and centrifugation; the recovered gelatin is dried, weighed and characterised.

4. Definitions & Key Terms

Collagen — the fibrous structural protein of hide, built from triple-helical chains rich in glycine, proline and hydroxyproline.

Gelatin — the water-soluble protein formed when collagen's triple helix is denatured (unwound) into single chains by acid/alkali treatment and heat.

Denaturation — loss of the ordered triple-helix structure (unfolding), without extensively breaking the peptide backbone.

Hydrolysis — breaking of peptide bonds by reaction with water (acid-catalysed here); excessive hydrolysis lowers molecular weight and gel strength.

Yield — the mass of dry gelatin recovered, expressed as a percentage of the dry weight of trimmings taken.

Liquid-to-solid ratio — the volume of extraction liquid (mL) per gram of dry trimmings — here 20:1, i.e. 20 mL acid per 1 g.

5. Group Allocation — Four Groups

All groups follow the same procedure and keep everything constant — 6% acetic acid, 20:1 liquid-to-solid ratio, 3 hours, 120-300 rpm — except the extraction temperature, which each group sets differently. Pooling the four results reveals the trade-off: lower temperatures tend to give stronger gel, while higher temperatures give more yield but can degrade quality.

Group	Assigned extraction temperature	Members
Group 1	60 °C (expect strong gel, lower yield)	_____
Group 2	70 °C	_____
Group 3	80 °C (SOP baseline)	_____
Group 4	90 °C (expect high yield, weaker gel)	_____

Keep the temperature within ± 2 °C of the assigned value throughout, and avoid boiling.

6. Materials & Chemicals

- Raw/ Delimed tannery trimmings — non-chrome, collagen-rich (raw material)
- Distilled water
- Glacial acetic acid ($\geq 99.5\%$)
- Sodium hydroxide (optional, for pH adjustment)

7. Equipment

- Analytical balance (± 0.001 g)
- Hot plate with magnetic stirrer and stir bar
- 250 mL conical flask; pre-weighed glass beakers
- Thermometer / digital temperature probe; water bath (optional)

- Whatman No. 1 filter paper; vacuum filtration setup (recommended)
- Centrifuge
- Drying oven; desiccator; muffle furnace
- pH meter; moisture analyser (optional)
- Stainless-steel scissors

8. Preparation of 6% (v/v) Acetic Acid

To make 100 mL: measure 6.0 mL of glacial acetic acid and add it slowly to 94 mL of distilled water (acid into water, never the reverse). Mix well. Prepare fresh before use.

Scale up as needed — each 1 g of trimmings needs 20 mL of this solution (20:1 ratio).

9. Sample Preparation

1. Collection: obtain fresh raw/ delimed trimmings free of chromium tanning chemicals.
2. Washing: wash thoroughly under running water, then rinse three times with distilled water.
3. Cutting: cut into small pieces of about 5–10 mm with stainless-steel scissors.
4. Drying: dry at 100 °C for 30 minutes (until surface moisture is gone); cool in a desiccator.

10. Extraction Procedure

1. Accurately weigh 1.000 ± 0.005 g of the dried trimmings and transfer to a 250 mL conical flask.
2. Add 20 mL of 6% acetic acid (the 20:1 ratio) so the trimmings are fully immersed. Measure and record the pH (expected 2.5–3.2).
3. Place on the magnetic stirrer and extract at YOUR assigned temperature (**60, 70, 80 or 90 °C**), 120–300 rpm, for 3 hours. Cover the flask with foil or a watch glass; hold the temperature within ± 2 °C; do not boil.
4. Cooling: remove the flask and let it cool naturally to 25–30 °C (about 20 minutes).
5. Filtration: filter through Whatman No. 1 paper; collect the filtrate in a centrifuge tube.
6. Centrifugation: centrifuge at 5000 rpm for 10 minutes at room temperature; carefully decant the clear supernatant and discard the sediment.
7. Drying: transfer the supernatant to a pre-weighed beaker and dry —
 Stage 1: 80 °C for 30 min;
 Stage 2: 100 °C for 30 min — or until constant weight (change < 0.01 g over 30 min).
 Cool in a desiccator and weigh immediately.

11. Yield Calculation

$$\text{Gelatin yield (\%)} = (W_{\gamma} / W_i) \times 100$$

where W_i = initial dry weight of trimmings and W_{γ} = recovered dry gelatin weight.

Worked example:

$$W_i = 1.000 \text{ g}, W_{\gamma} = 0.315 \text{ g} \rightarrow \text{yield} = (0.315 / 1.000) \times 100 = 31.5\%.$$

12. Characterization of the Gelatin

Determine the following properties of your recovered gelatin (do the core ones; the optional ones may be shared across groups or demonstrated).

Yield (%) — gravimetric, from the formula above (core).

13. Observations & Data Recording

A. Your own group's data

Parameter	Value
Sample ID	
Assigned temperature (°C)	
Initial dry weight, W_i (g)	
Acid concentration (%) / volume (mL)	
pH before extraction	
Extraction time (h) / stirring (rpm)	
Final gelatin weight, W_f (g)	
Yield (%)	
pH (1% solution)	
Appearance / colour	

B. Pooled class results (temperature series — copy every group's result)

Temperature	Yield (%)	Appearance
60 °C		
70 °C		
80 °C		
90 °C		

Your working

14. Quality Control

- Hold the extraction temperature within ± 2 °C of your assigned value.
- Measure and record the pH before extraction (expected 2.5–3.2).
- Dry gelatin to constant weight before weighing (change < 0.01 g over 30 min).
- Run a minimum of three independent extractions ($n = 3$) where possible and average.
- Good gelatin is light yellow to pale cream, homogeneous, free of burnt particles and visible impurities.

15. Expected Results

Parameter	Typical range
Yield (dry basis)	20 – 40%
Appearance	light yellow, transparent when dissolved

16. Safety Precautions

- Wear a lab coat, gloves and safety goggles.
- Always add acid to water, never water to concentrated acid.
- Use heat-resistant gloves for hot glassware; do not let the extraction boil.
- Balance centrifuge tubes in pairs before spinning.
- Use only non-chrome (delimed) trimmings — chrome-bearing material must never be used for gelatin.
- Dispose of acidic waste per institutional chemical-waste procedures.